Transformations of Quadratics in Vertex Form

Comprehensive Notes and Practice

1 Core Concept: Vertex Form

The **vertex form** of a quadratic function is:

$$\boxed{y = a(x-h)^2 + k} \tag{1}$$

Where:

- (h, k) is the **vertex** of the parabola
- a controls the **direction** and **width** of the parabola

2 Transformation Examples

Worked Examples	
Example 1: $y = 2(x-3)^2 + 1$	**Example 2:** $y = -\frac{1}{2}(x+4)^2 - 2$
Starting from $y = x^2$: 1. **Vertical stretch by 2** ($a = 2$) $\rightarrow$ Narrower parabola 2. **Shift right 3 units** ($h = 3$) $\rightarrow$ Vertex: $(0,0) \rightarrow (3,0)$ 3. **Shift up 1 unit** ($k = 1$) $\rightarrow$ Vertex: $(3,0) \rightarrow (3,1)$ **Final vertex:** $(3,1)$ **Opens:** Upward ($a > 0$)	Starting from $y = x^2$: 1. **Vertical compression by** $\frac{1}{2}$ ($\lvert a \rvert = \frac{1}{2}$) $\rightarrow$ Wider parabola 2. **Reflection over x-axis** ($a < 0$) $\rightarrow$ Opens downward 3. **Shift left 4 units** ($h = -4$) $\rightarrow$ Vertex: $(0,0) \rightarrow (-4,0)$ 4. **Shift down 2 units** ($k = -2$) $\rightarrow$ Vertex: $(-4,0) \rightarrow (-4,-2)$ **Final vertex:** $(-4,-2)$ **Opens:** Downward ($a < 0$)

Transformation	**Effect**	**Example**
$a > 1$	Narrower (vertical stretch)	$y = 3x^2$
$0 < a < 1$	Wider (vertical compression)	$y = 0.25x^2$
$a < 0$	Opens down (reflection)	$y = -x^2$
$(x - h)$, $h > 0$	Shift right h units	$y = (x - 5)^2$
$(x + h)$, $h > 0$	Shift left h units	$y = (x + 5)^2$
$+k$	Shift up k units	$y = x^2 + 7$
$-k$	Shift down k units	$y = x^2 - 3$

3 Practice Problems

3.1 Section A: Identifying Transformations

For each function, identify the vertex, direction (up/down), and describe all transformations from $y = x^2$.

1. $y = 3(x - 2)^2 + 5$

Answer:

Vertex: ______________ Opens: ______________

2. $y = -(x + 1)^2 - 4$

Answer:

Vertex: ______________ Opens: ______________

3. $y = \frac{1}{2}(x - 6)^2$

Answer:

Vertex: ______________ Opens: ______________

4. $y = -2(x + 3)^2 + 1$

Answer:

Vertex: ______________ Opens: ______________

5. $y = (x+5)^2 - 7$

Answer:

Vertex: ________ Opens: ________

6. $y = 4(x-1)^2 - 2$

Answer:

Vertex: ________ Opens: ________

7. $y = -\frac{1}{4}(x+2)^2 + 3$

Answer:

Vertex: ________ Opens: ________

8. $y = -(x-4)^2$

Answer:

Vertex: ________ Opens: ________

3.2 Section B: Writing Equations from Descriptions

Write the vertex form equation for each description.

9. Vertex at $(2, 3)$, opens upward, vertical stretch by factor of 2

Answer:

Answer: ________

10. Vertex at $(-1, -5)$, opens downward, no vertical stretch/compression

Answer:

Answer: ________________

11. Vertex at $(0, 4)$, opens upward, vertical compression by factor of $\frac{1}{3}$

Answer:

Answer: ________________

12. Vertex at $(7, -2)$, opens downward, vertical stretch by factor of 5

Answer:

Answer: ________________

13. The parent function $y = x^2$ shifted left 3 units and down 6 units

Answer:

Answer: ________________

3.3 Section D: Graphing Practice

Sketch graphs for these functions. Label the vertex and at least 2 other points.

14. $y = (x-2)^2 + 1$

15. $y = -\frac{1}{2}(x+1)^2 + 3$

16. $y = 2(x-3)^2 - 4$

17. $y = -(x+2)^2 - 1$

3.4 Section E: Advanced Application

18. A parabola has vertex $(-2, 5)$ and passes through the point $(0, 1)$. Write its equation in vertex form.

Answer:

19. Write an equation for a parabola that opens downward, is wider than $y = x^2$, and has its vertex at $(3, -2)$.

5cm

Answer:

20. If the vertex of $y = a(x-2)^2 + 3$ is at the point $(2, 3)$, and the parabola passes through $(4, 7)$, find the value of a.

Answer: $a =$

21. A parabola in vertex form $y = a(x-1)^2 - 8$ passes through the point $(3, 4)$. Find the value of a.

Answer: $a =$

22. The parabola $y = a(x+5)^2 + k$ has a vertex at $(-5, 7)$ and passes through the point $(-3, 3)$. Find the values of a and k.

Answer

$a =$ ________________ $k =$ ________________